Welcome!

DSC 340 Applied Machine Learning

5 February 2025

About this Course. What is Data Science? The Data Science Lifecycle.

The Plan

- Welcome!
- Introductions
- Course Overview
- What is Data Science?
- What is Applied Machine Learning?
- Data Science Lifecycle (first look)

Next 2 days: Example data analytics problem (linear regression)

About DSC 340

Syllabus

About the Course (Note: formerly DSC 305)

- Tools
- Workflow (Data Analytics Lifecycle)
- Assess
- Assess and draw conclusions
- Communicate
  - Orally
  - In writing
  - Visualization
- Hands-on experience with varied datasets

Before this course

Prerequisites:

- DSC 230 — Statistics foundation, linear regression
  - MAT 130 Intro stats
- DSC 270 — Python, NumPy, Pandas, SQL, Excel, etc.
  - CSC 170 Intro coding
- CSC 165 — Introductory calculus (minimizing a function)

- Or see instructor ASAP!

After this Course

Learning outcomes:

- What a data scientist does
- Apply data analytics lifecycle to a problem
- Tools: Py3, Jupyter, pandas, sklearn, Matplotlib, tf? …
- Jupyter = Code + Output + Markdown (browser)
- Data sets

After this Course (cont.)

- Exploratory data analysis
- Prep data for model building
- Methods (e.g., logistic regression)
- Model data appropriately (e.g., scale)
- Occam's razor
- Analyze and assess results
- Present results orally and writing (+ data viz)
- Experience: labs + significant data analytics project

Resources

- Textbooks: EMC, PDSH
- Many (!) online resources
- Jupyter Notebook (Olin 208, Anaconda, CoLab)
- Office Hours: Weekdays at 3:00 PM (for now)
- Disabilities, Library, ESL, Computer Labs (Olin 208, etc.)
- Coding Collaborators (Partners): P2-P3 and P4-P5

Grades

• Data Analytics Projects ("Labs") (about 5 @ 5%)	25%
• Final Project	8%
• Tests (about 6, usu. on Wednesdays)	54%
• Comprehensive Final Exam	13%

	100%

Workflow and Late Submissions

- Attendance
- No retakes*
- No late submissions*
- Academic honesty
- Class participation
- Respect for others
- Email, Moodle

*See syllabus for discussion of special cases

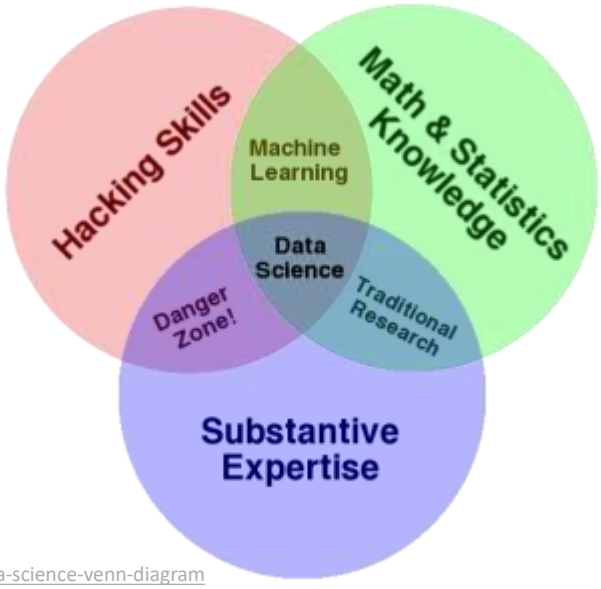
Questions?

- What are your questions about this course?

What is Data Science?

EMC 1, PDSH Preface, Other Resources (see Moodle)

Data Science



http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagram



https://www.edureka.co/blog/what-is-big-data/



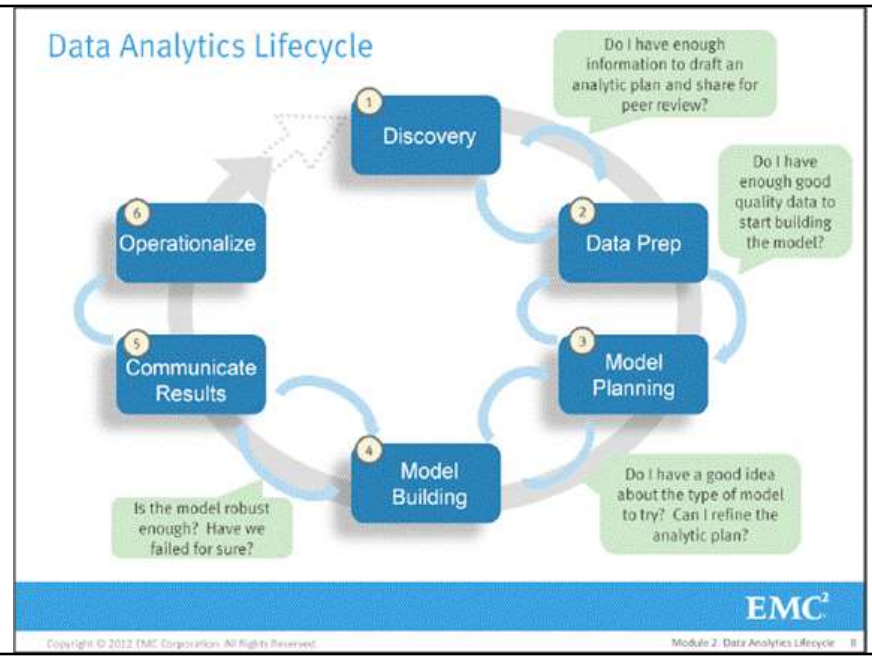
https://medium.com/@tomaszdudek/but-what-is-this-machine-learning-engineer-actually-doing-18464d5c699

Data Analytics Roles

- **Organization User:** understands domain area, benefits from work
- **Project Sponsor:** high-level direction from administration
- **Project Manager:** supervises process to meet milestones/objectives
- **BI Analyst:** Business domain expertise, dashboards and reports
- **DB Administrator:** IT support for DB: roles, permissions, installations
- **Data Engineer:** Programming, complex SQL queries,
- **Data Scientist:** The expert in the room for techniques, models, and evaluating result: Choose the right method for the problem at hand.

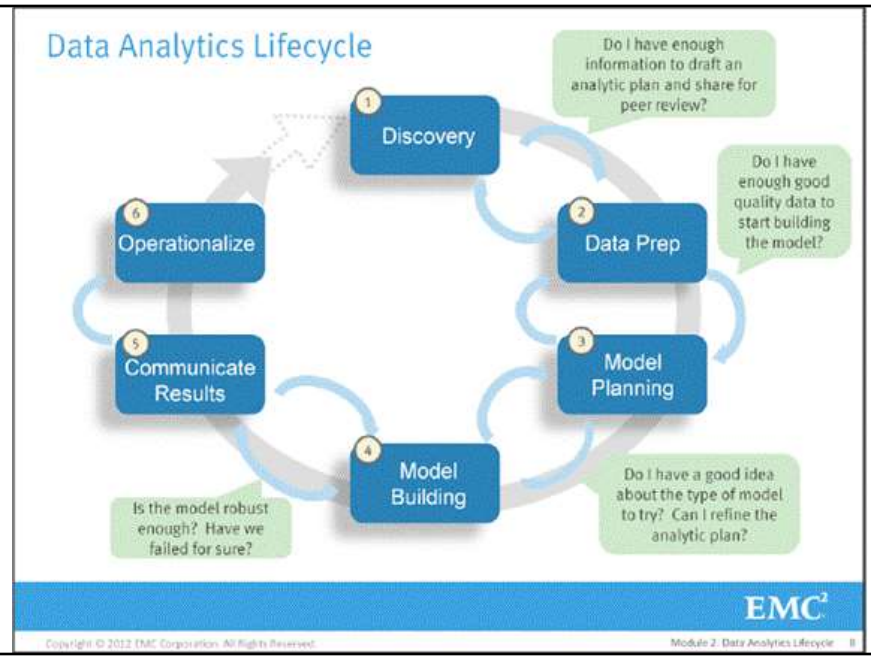
Data Analytics Lifecycle

EMC 2


Data Analytics Lifecycle
Do I have enough information to draft an analytic plan and share for peer review?
1 Discovery
2 Data Prep
Do I have enough good quality data to start building the model?
3 Model Planning
Do I have a good idea about the type of model to try? Can I refine the analytic plan?
4 Model Building
Is the model robust enough? Have we failed for sure?
5 Communicate Results
6 Operationalize
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Module 2: Data Analytics Lifecycle 8

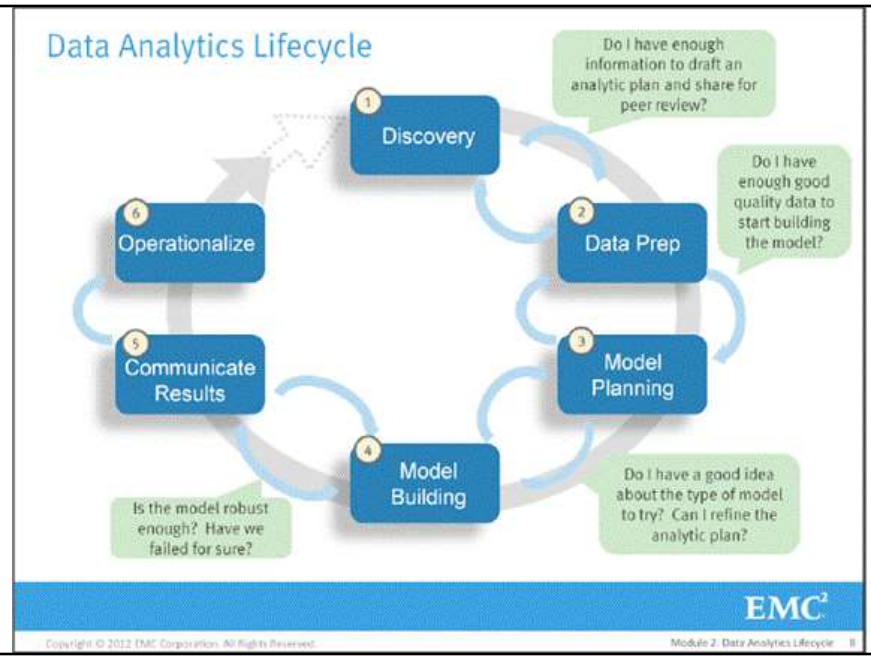
1. Discovery

- Learn the domain (business, science, etc.)
- What resources do we have? (people, tech, tools, data?)
- What problem are we solving? (frame the problem)
- Who are the stakeholders? (Who is affected? What are the risks?)
- Interview the Analytics Sponsor
- Developing initial hypotheses
- Identifying potential data sources


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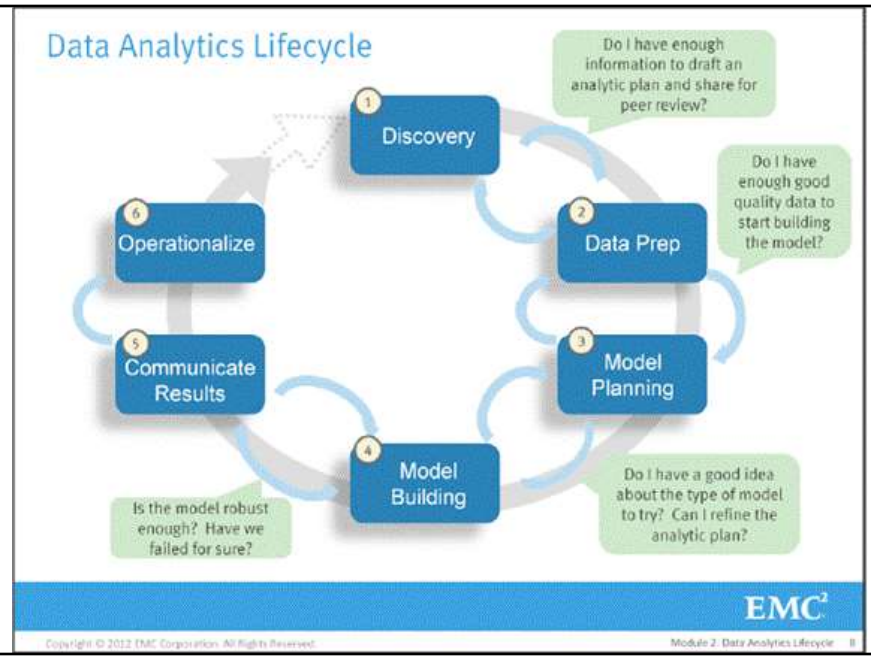
2. Data Preparation

- Sandbox data (no live databases!) analytic vs. operational DB
- Make sure you have the resources (data, tech) you need
- Learn about the data (semantics, missing values, errors, duplicates …)
- Data conditioning (address issues discovered, structure)
- Survey and visualize (is it representative? Consistent? Time frames?)


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3. Model Planning

- Data exploration and variable selection
  - Already cleaned, but need to understand relationship between variables
- Model selection:
  - Is this a classification problem? Clustering?
  - Which method(s) are most appropriate?

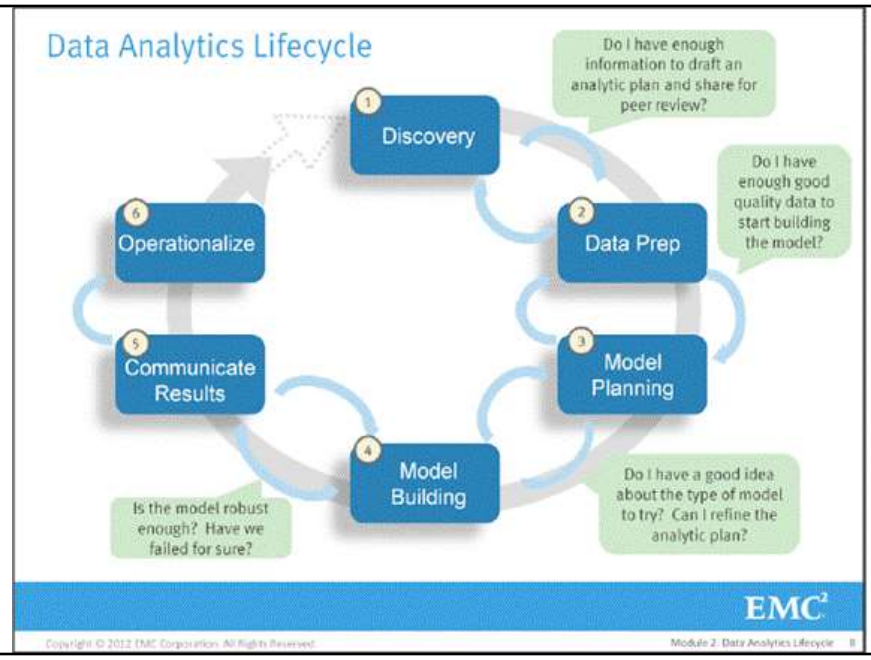

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4. Model Building

- "Train" model ("fit" data)
- Does the model appear valid and accurate?
- Does the output/behavior make sense to domain experts?
- Do the parameter values of the fitted model make sense in domain?
- Is the model sufficiently accurate to meet the goal?
- Does it avoid intolerable mistakes? (depends on context!)
  - For example: false positives. How bad?

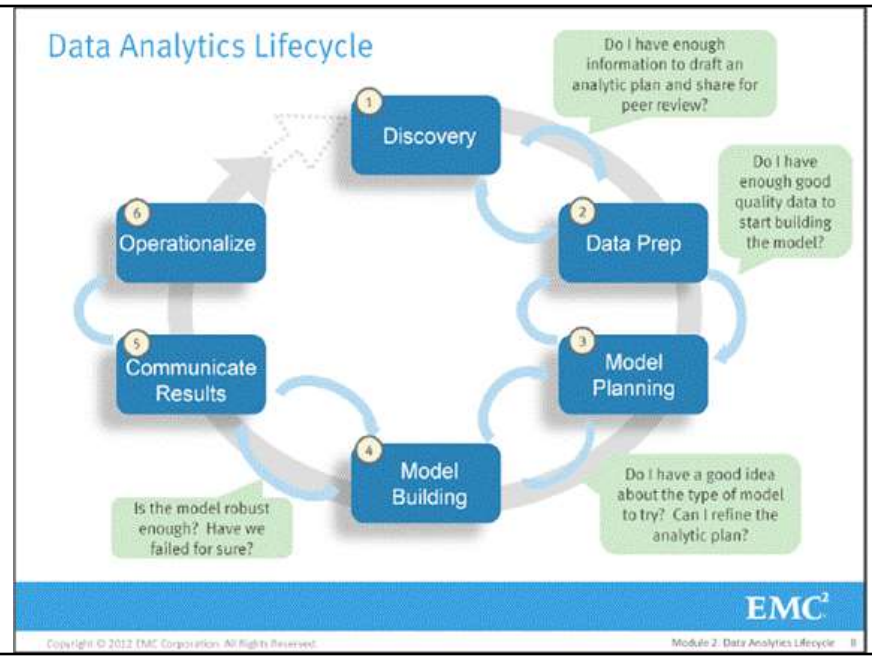
4. Model Building (cont.)

- Do we need more data? Transformations? Less data?
- Will the model support runtime (operational) requirements?
- Is a different form of the model needed for operationalization?


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5. Communicate Results

- Did we succeed or fail?
- Common sense evaluation
- Present key results
- Make recommendations for follow-up
- Document work of team


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